test

March 17, 2026

[ ]:
```
import pandas as pd
import matplotlib.pyplot as plt
import os

plot_configs = [
    {
        'file': 'LED-Azerostart.csv',
        'x': 'Hours',
        'y': 'Relative Change',
        'group': 'Unit Number',
        'title': 'LED Degradation Paths (Relative Change)'
    },
    {
        'file': 'GaAsLaser.csv',
        'x': 'Hours',
        'y': 'Percent Increase',
        'group': 'Unit Number',
        'title': 'GaAs Laser Threshold Current Increase'
    },
    {
        'file': 'DiskBer.csv',
        'x': 'Hours',
        'y': 'Byte Error Rate',
        'group': 'Disk Number',
        'title': 'Disk BER Degradation (Byte Error Rate)'
    },
    {
        'file': 'MetalWear.csv',
        'x': 'Cycles',
        'y': 'Microns',
        'group': 'Unit Number',
        'title': 'Metal Wear Degradation (Microns)'
    },
    {
        'file': 'Resistor.csv',
        'x': 'Thousands of Hours',
        'y': 'Percent Increase',
```

```
        'group': 'Resistor ID',
        'title': 'Resistor Resistance Growth'
    },
    {
        'file': 'AlloyA.csv',
        'x': 'Millions of Cycles',
        'y': 'Length (in)',
        'group': 'Specimen Number',
        'title': 'Alloy A Crack Growth (Length)'
    },
    {
        'file': 'DeviceB.csv',
        'x': 'Hours',
        'y': 'Power Drop (dB)',
        'group': 'Device Number',
        'title': 'Device B Power Drop'
    }
]

def plot_all_degradation(configs):
    for cfg in configs:
        file_name = cfg['file']
        if not os.path.exists(file_name):
            print(f"   : {file_name}")
            continue

        print(f"  : {file_name}...")
        df = pd.read_csv(file_name)

        plt.figure(figsize=(10, 6))

        units = df[cfg['group']].unique()

        for unit in units:
            unit_data = df[df[cfg['group']] == unit]
            unit_data = unit_data.sort_values(by=cfg['x'])

            plt.plot(unit_data[cfg['x']], unit_data[cfg['y']],
                     marker='o', markersize=3, alpha=0.6,
                     label=f'Unit {unit}' if len(units) <= 10 else "")

        plt.xlabel(cfg['x'], fontsize=12)
        plt.ylabel(cfg['y'], fontsize=12)
        plt.title(cfg['title'], fontsize=14)
        plt.grid(True, linestyle='--', alpha=0.5)

        if 0 < len(units) <= 10:
```

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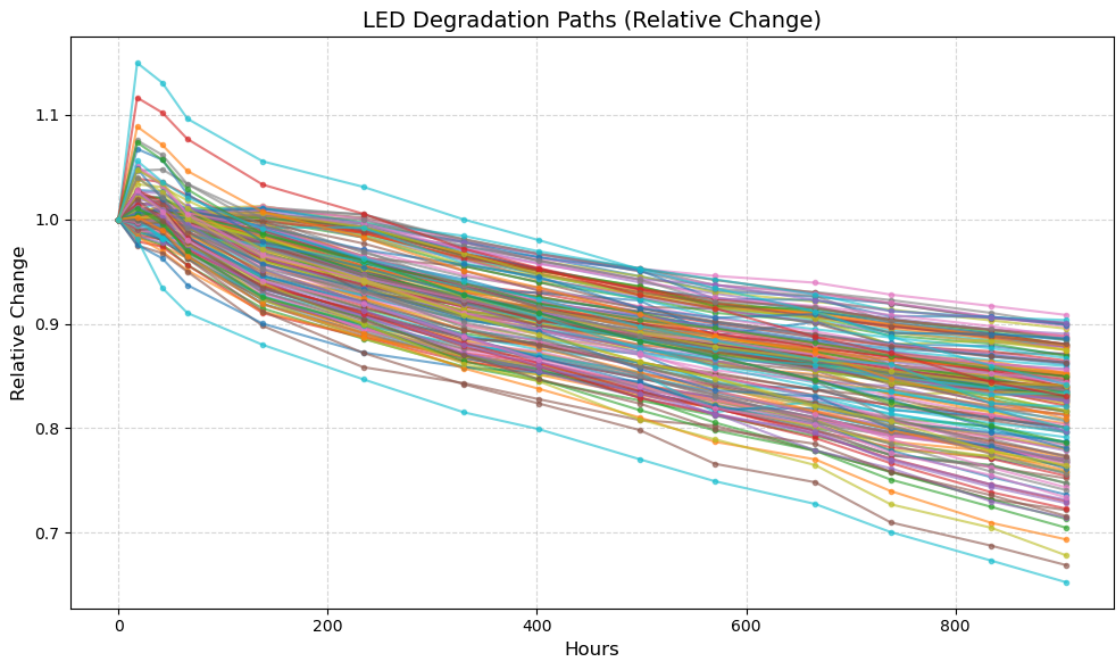
            plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

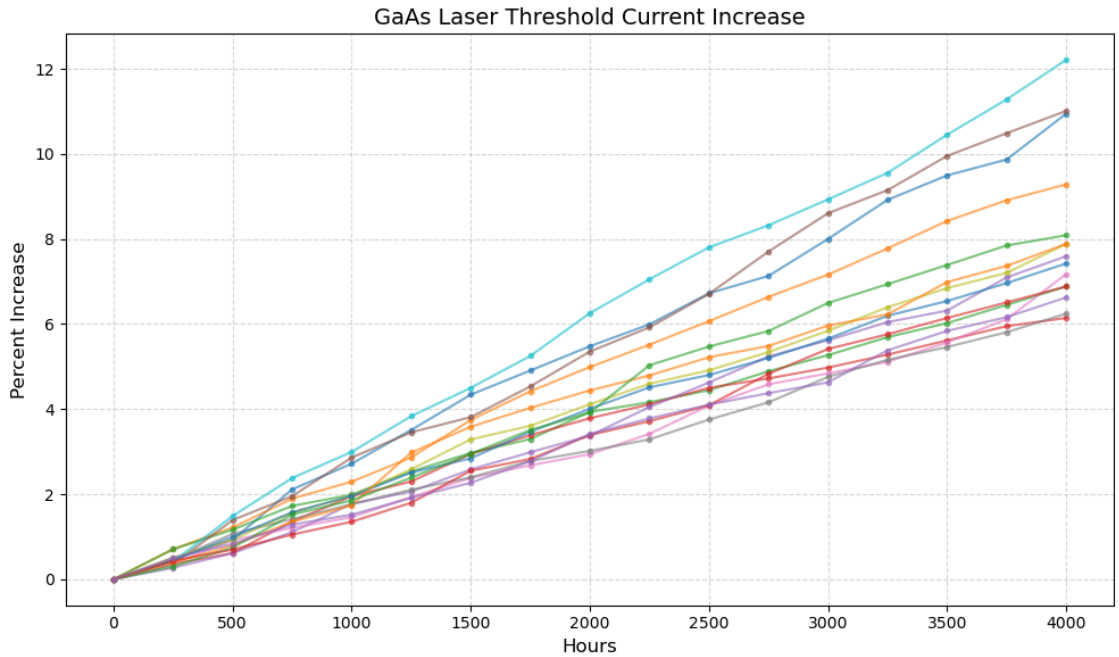
        plt.tight_layout()
        plt.show()

plot_all_degradation(plot_configs)
```

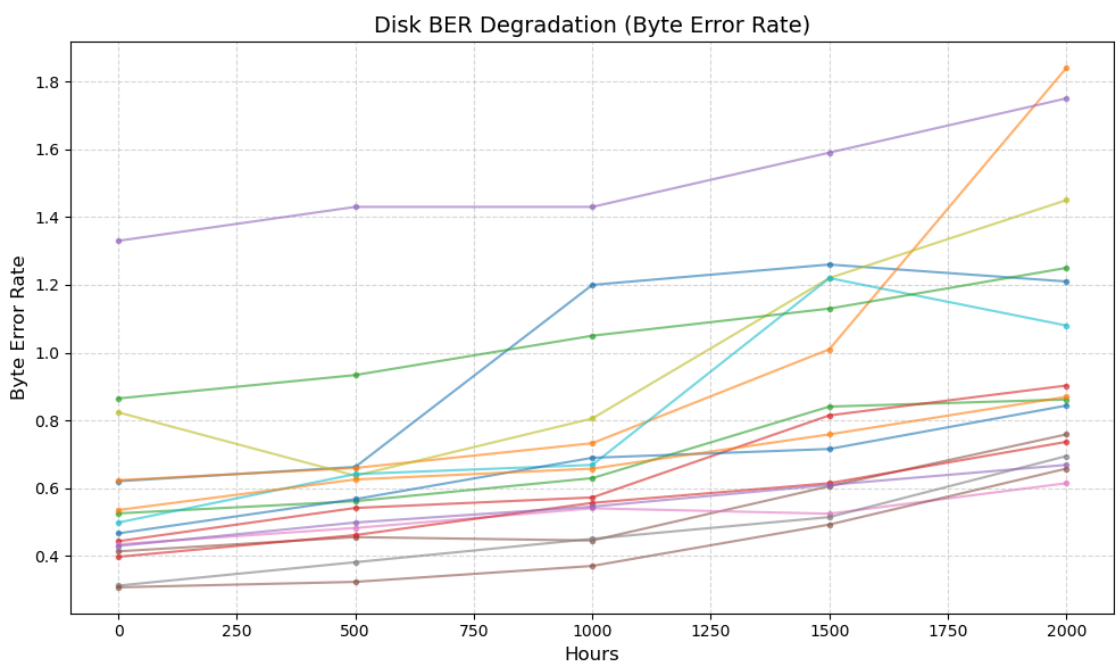
: LED-Azerostart.csv…



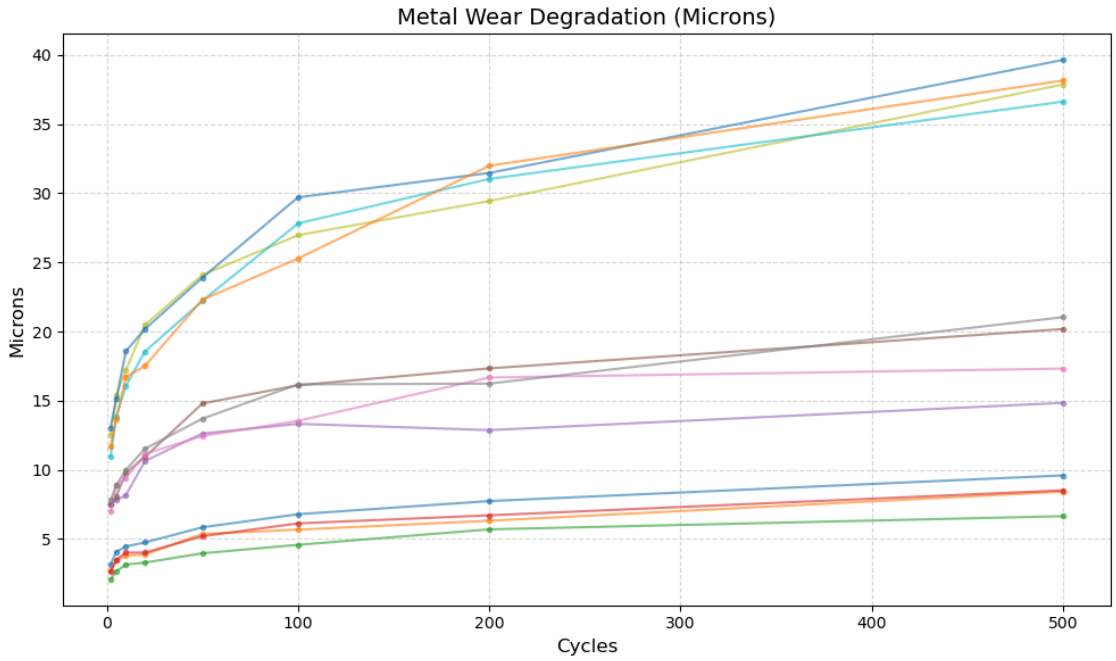
: GaAsLaser.csv…



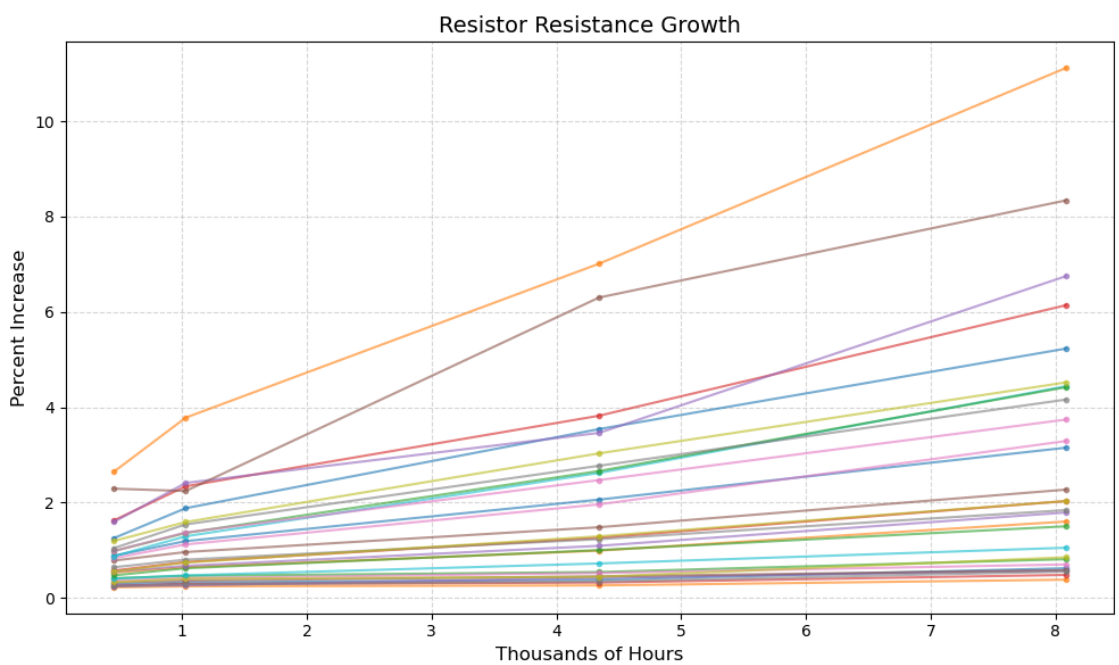
: `DiskBer.csv…`



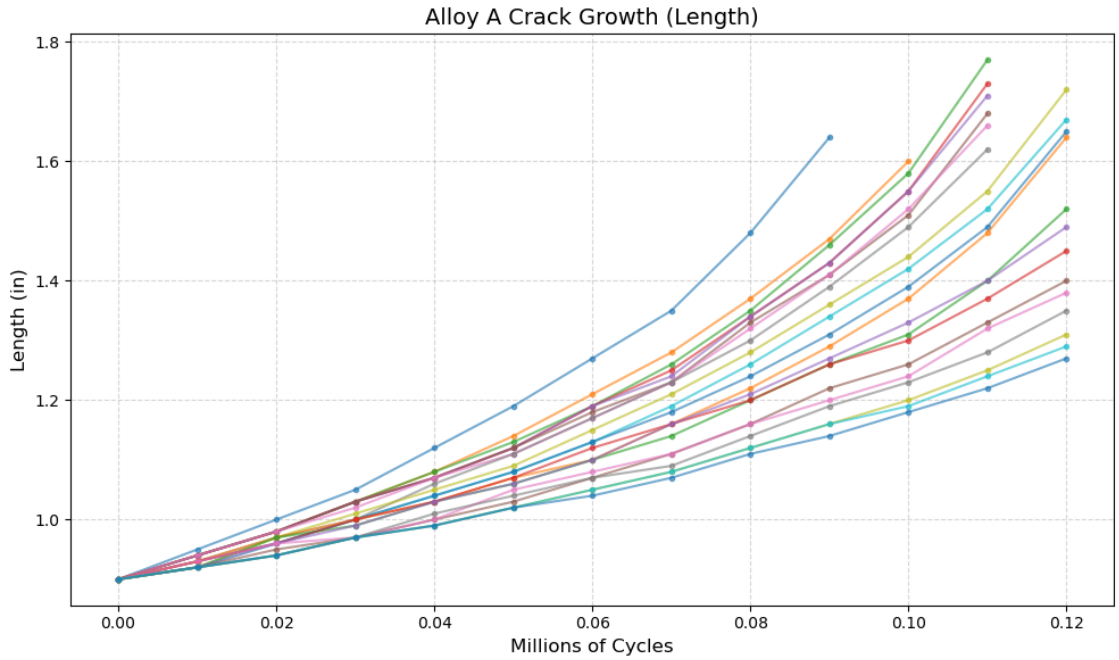
: `MetalWear.csv…`



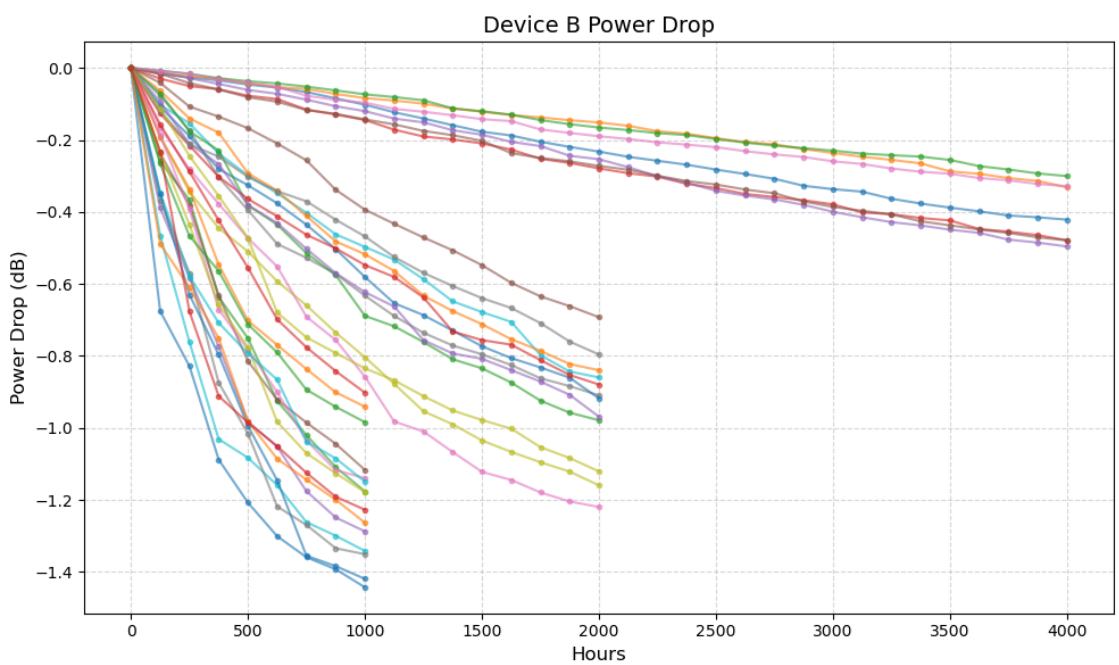
: `Resistor.csv…`



: `AlloyA.csv…`


Alloy A Crack Growth (Length)
Length (in)
Millions of Cycles
1.8
1.6
1.4
1.2
1.0
0.00
0.02
0.04
0.06
0.08
0.10
0.12

: DeviceB.csv…


Device B Power Drop
Power Drop (dB)
Hours
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
−1.4
0
500
1000
1500
2000
2500
3000
3500
4000

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[ ]:
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